2026 05 28
발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

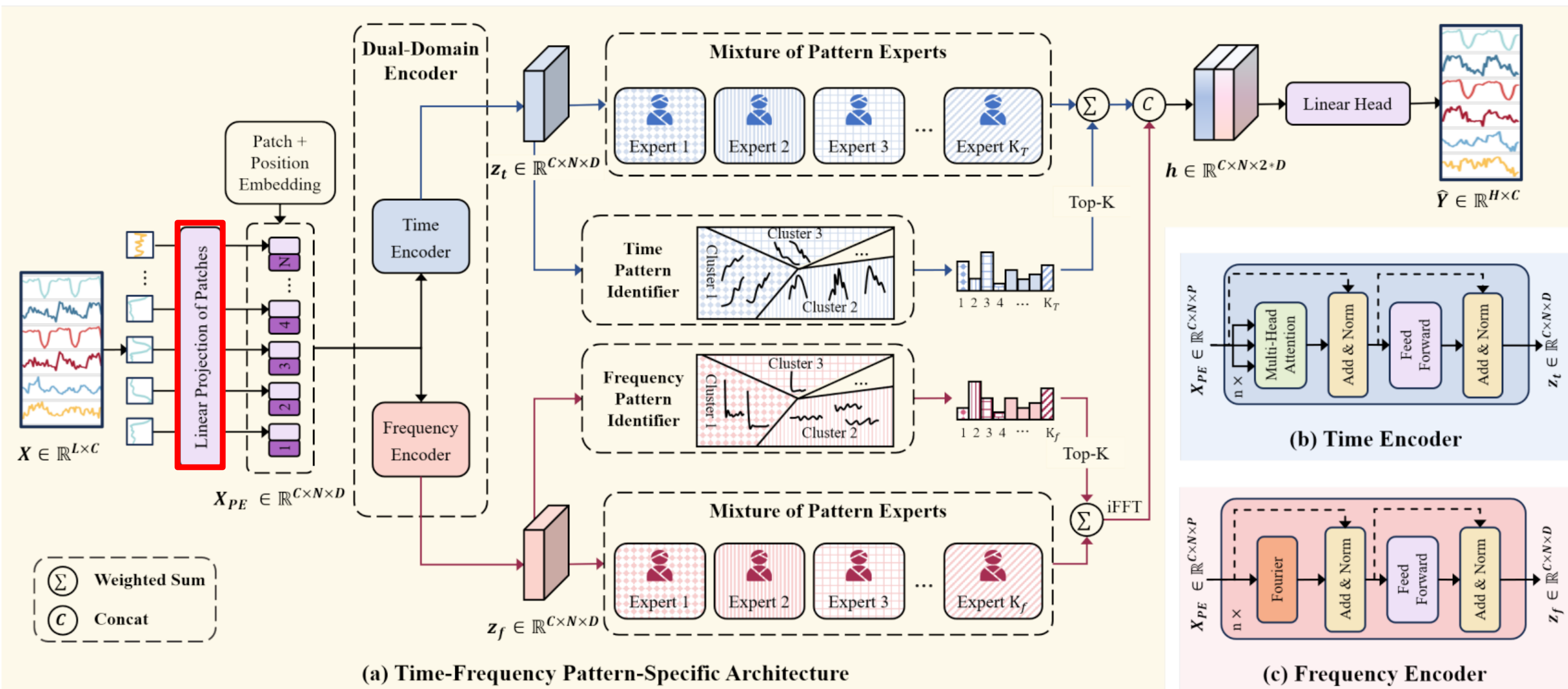
KWANGWOON
UNIVERSITY

이번 주 진행사항

- Time-Frequency Pattern-Specific (TFPS)
  - 아이디어 적용 실험
- TimeEmb
  - 논문 리뷰

아이디어

FAIR Lab



- Linear Projection of Patches
  - 기존의 고정된 Patch 길이 대신 8, 16, 32와 같이 여러 Scale로 설정하여 병렬로 처리하고, concat을 통해 각 특징들을 합한 뒤 시계열 데이터의 장·단기 패턴을 한 번에 학습

실험 세팅

- 사용한 모델: TFPS
- 재현 실험 데이터셋: ETTh1, ETTm1

Experiment	ETTh1
Learning rate	5×10^{-4}
Epoch	100
Batch size	128
Loss function	MSE Loss
Seq_len	96
Pred_len	96/192/336/720
Layers	3
Expert	4~16

재현 실험 (ETTh1)

	ETTh1 Reproduction		ETTh1 Reproduction + Multi-Scale Patch	
Prediction length	MSE	MAE	MSE	MAE
96	**0.397**	<u>0.413</u>	**0.397**	**0.412**
192	**0.455**	**0.445**	<u>0.457</u>	<u>0.448</u>
336	**0.488**	<u>0.469</u>	<u>0.489</u>	**0.460**
720	**0.486**	**0.476**	<u>0.497</u>	<u>0.477</u>

재현 실험 (ETTm1)

	ETTm1 Reproduction		ETTm1 Reproduction + Multi-Scale Patch	
Prediction length	MSE	MAE	MSE	MAE
96	0.327	0.369	**0.326**	**0.367**
192	0.378	0.396	**0.376**	**0.394**
336	0.412	0.417	**0.411**	**0.415**
720	0.486	**0.462**	**0.484**	0.465

TimeEmb

TimeEmb: A Lightweight Static-Dynamic Disentanglement Framework for Time Series Forecasting

Mingyuan Xia*
Jilin University
xiamy2322@mails.jlu.edu.cn

Chunxu Zhang*
Jilin University
zhangchunxu@jlu.edu.cn

Zijian Zhang†
Jilin University
zhangzijian@jlu.edu.cn

Hao Miao
The Hong Kong Polytechnic University
hao.miao@polyu.edu.hk

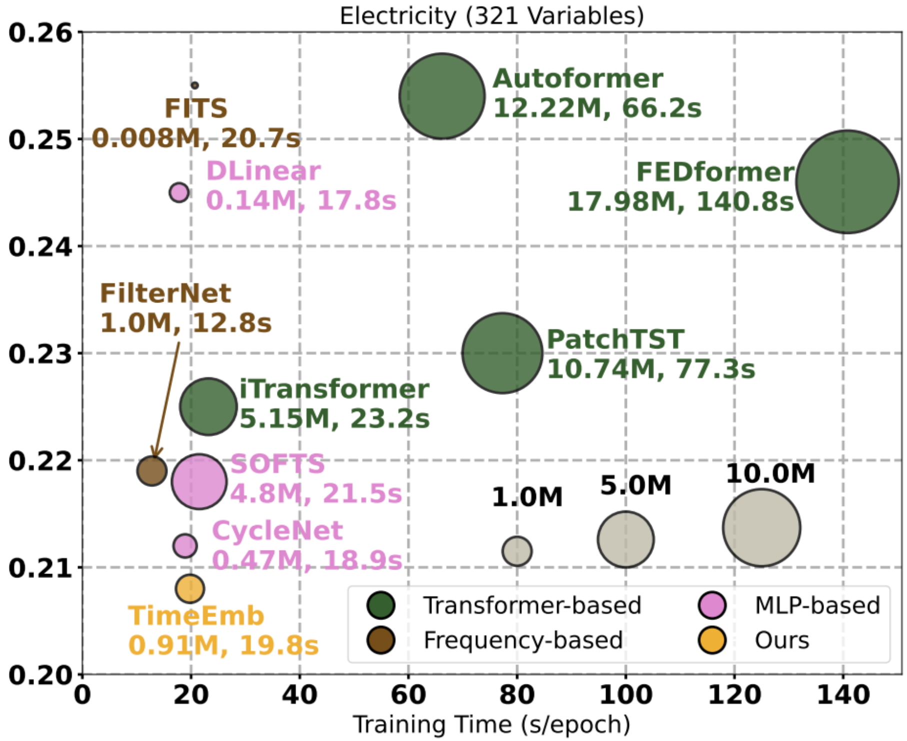
Qidong Liu
Xi'an Jiaotong University
liuqidong@xjtu.edu.cn

Yuanshao Zhu
City University of Hong Kong
yuanshao@ieee.org

Bo Yang
Jilin University
ybo@jlu.edu.cn

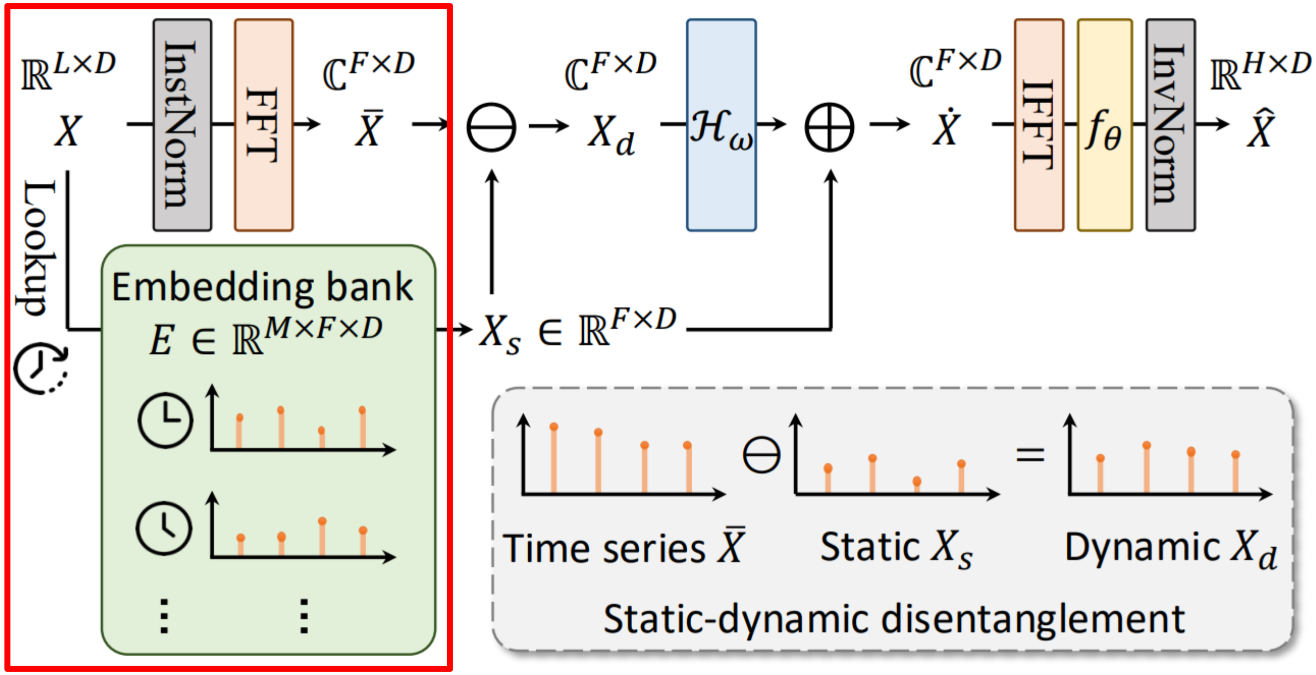
- arXiv 등록일: 2025년 10월 1일
- 인용 수: 11회(Google Scholar, 2026-05-26)
- Published at NeurIPS 2025
- 링크: https://openreview.net/pdf?id=sLfMvrkn6T

Background



- Transformer 기반 모델은 Self-Attention의 이차 복잡도로 인해 높은 파라미터 수와 연산 비용 발생, MLP 기반 모델은 단순 구조로 효율성을 높였으나 시계열의 복잡한 패턴을 완벽히 포착하지 못함
- 위 문제를 해결하기 위해 TimeEmb를 통해 시계열을 정적 성분과 동적 성분으로 분해하여 처리함

Overview



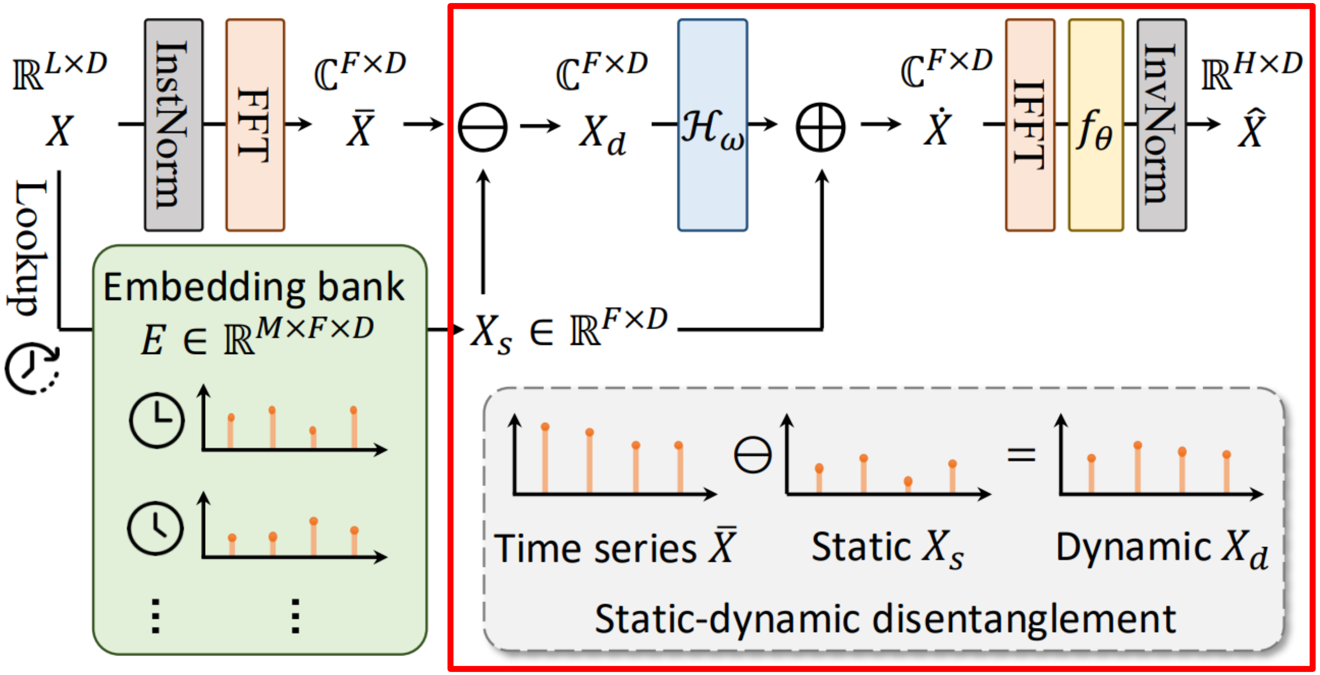
- Domain Transformation
  - 입력 데이터 X를 InstNorm을 통해 정규화, FFT로 Frequency Representation $\bar{X} \in \mathbb{C}^{\mathbb{F}\times\mathbb{D}}$ 생성

    $\bar{X}[k] = \sum_{n=0}^{L-1} X[n]e^{-j2\pi kn/L}, \quad k = 0,1,\ldots,F-1$
- Lookup & Embedding bank
  - 입력 데이터 X의 마지막 타임스탬프를 기반으로 주기 내 위치 계산 $t_{last}(mod\ M)$

    학습 가능한 Embedding bank에서 해당 시간대에 고정된 시간 정적 성분 X_s를 조회하여 추출
  - Backpropagation으로 전체 데이터셋의 주파수 구조를 학습

$\bar{X}$: 주파수 표현인 복소수 행렬

Overview



- Static-dynamic disentanglement
  - Frequency Representation $\bar{X}$에서 정적 성분 X_s를 빼서 동적 성분 X_d를 구함
  - 주파수 필터링($\mathcal{H}_\omega$)을 통해 학습 가능한 주파수 필터 가중치를 요소별로 곱하고, 정적 성분 X_s와 더하여 $\dot{X}$ 생성
- Prediction Layer
  - $\dot{X}$을 IFFT를 통해 시간 도메인으로 변환하고, f_θ으로 예측값 생성
  - 이후 예측값에 InvNorm을 적용해 역정규화하여 원래 스케일로 변환

Performance comparison

FAIR Lab

KWANGWOON UNIVERSITY

Model		TimeEmb (ours)		CycleNet 2024		Fredformer 2024		FilterNet 2024		iTransformer 2024		PatchTST 2023		FITS 2024		FreTS 2023		DLinear 2023	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.366±0.001**	**0.387±0.001**	0.378	0.391	0.373	0.392	0.375	0.394	0.386	0.405	0.394	0.406	0.386	0.396	0.395	0.407	0.386	0.400
	192	**0.417±0.001**	**0.416±0.001**	0.426	0.419	0.433	0.420	0.436	0.422	0.441	0.436	0.440	0.435	0.436	0.423	0.448	0.440	0.437	0.432
	336	**0.457±0.001**	**0.436±0.001**	0.464	0.439	0.470	0.437	0.476	0.443	0.487	0.458	0.491	0.462	0.478	0.444	0.499	0.472	0.481	0.459
	720	**0.459±0.002**	0.460±0.001	0.461	0.460	0.467	**0.456**	0.474	0.469	0.503	0.491	0.487	0.479	0.502	0.495	0.558	0.532	0.519	0.516
	avg	**0.425±0.001**	**0.425±0.001**	0.432	0.427	0.435	0.426	0.440	0.432	0.454	0.447	0.453	0.446	0.451	0.440	0.475	0.463	0.456	0.452
ETTh2	96	**0.277±0.001**	**0.328±0.001**	0.285	0.335	0.293	0.342	0.292	0.343	0.297	0.349	0.288	0.340	0.295	0.350	0.309	0.364	0.333	0.387
	192	**0.356±0.001**	**0.379±0.001**	0.373	0.391	0.371	0.389	0.369	0.395	0.380	0.400	0.376	0.395	0.381	0.396	0.395	0.425	0.477	0.476
	336	0.400±0.002	0.417±0.001	0.421	0.433	**0.382**	**0.409**	0.420	0.432	0.428	0.432	0.440	0.451	0.426	0.438	0.462	0.467	0.594	0.541
	720	**0.416±0.001**	0.437±0.002	0.453	0.458	**0.415**	**0.434**	0.430	0.446	0.427	0.445	0.436	0.453	0.431	0.446	0.721	0.604	0.831	0.657
	avg	**0.362±0.001**	**0.390±0.001**	0.383	0.404	0.365	0.393	0.378	0.404	0.383	0.407	0.385	0.410	0.383	0.408	0.472	0.465	0.559	0.515
ETTm1	96	**0.304±0.001**	**0.343±0.001**	0.319	0.360	0.326	0.361	0.318	0.358	0.334	0.368	0.329	0.365	0.355	0.375	0.335	0.372	0.345	0.372
	192	**0.354±0.001**	**0.373±0.001**	0.360	0.381	0.363	0.380	0.364	0.383	0.377	0.391	0.380	0.394	0.392	0.393	0.388	0.401	0.380	0.389
	336	**0.379±0.001**	**0.393±0.001**	0.389	0.403	0.395	0.403	0.396	0.406	0.426	0.420	0.400	0.410	0.424	0.414	0.421	0.426	0.413	0.413
	720	**0.435±0.001**	**0.428±0.001**	0.447	0.441	0.453	0.438	0.456	0.444	0.491	0.459	0.475	0.453	0.487	0.449	0.486	0.465	0.474	0.453
	avg	**0.368±0.001**	**0.384±0.001**	0.379	0.396	0.384	0.395	0.384	0.398	0.407	0.410	0.396	0.406	0.415	0.408	0.408	0.416	0.403	0.407
ETTm2	96	**0.163±0.001**	**0.242±0.001**	**0.163**	0.246	0.177	0.259	0.174	0.257	0.180	0.264	0.184	0.264	0.183	0.266	0.189	0.277	0.193	0.292
	192	**0.226±0.001**	**0.285±0.001**	0.229	0.290	0.243	0.301	0.240	0.300	0.250	0.309	0.246	0.306	0.247	0.305	0.258	0.326	0.284	0.362
	336	0.286±0.001	**0.324±0.001**	**0.284**	0.327	0.302	0.340	0.297	0.339	0.311	0.348	0.308	0.346	0.307	0.342	0.343	0.390	0.369	0.427
	720	**0.383±0.001**	**0.381±0.001**	0.389	0.391	0.397	0.396	0.392	0.393	0.412	0.407	0.409	0.402	0.407	0.399	0.495	0.480	0.554	0.522
	avg	**0.265±0.001**	**0.308±0.001**	0.266	0.314	0.279	0.324	0.276	0.322	0.288	0.332	0.287	0.330	0.286	0.328	0.321	0.368	0.350	0.401
Weather	96	**0.150±0.001**	**0.190±0.001**	0.158	0.203	0.163	0.207	0.162	0.207	0.174	0.214	0.176	0.217	0.166	0.213	0.174	0.208	0.196	0.255
	192	**0.200±0.001**	**0.238±0.001**	0.207	0.247	0.211	0.251	0.210	0.250	0.221	0.254	0.221	0.256	0.213	0.254	0.219	0.250	0.237	0.296
	336	**0.259±0.001**	**0.282±0.001**	0.262	0.289	0.267	0.292	0.265	0.290	0.278	0.296	0.275	0.296	0.269	0.294	0.273	0.290	0.283	0.335
	720	0.339±0.001	0.336±0.001	0.344	0.344	0.343	0.341	0.342	0.340	0.358	0.347	0.352	0.346	0.346	0.343	**0.334**	**0.332**	0.345	0.381
	avg	**0.237±0.001**	**0.262±0.001**	0.243	0.271	0.246	0.272	0.245	0.272	0.258	0.278	0.256	0.279	0.249	0.276	0.250	0.270	0.265	0.317
Electricity	96	**0.136±0.001**	0.231±0.001	**0.136**	**0.229**	0.147	0.241	0.147	0.245	0.148	0.240	0.164	0.251	0.200	0.278	0.176	0.258	0.197	0.282
	192	0.153±0.001	0.246±0.001	**0.152**	**0.244**	0.165	0.258	0.160	0.250	0.162	0.253	0.173	0.262	0.200	0.280	0.175	0.262	0.196	0.285
	336	**0.170±0.001**	**0.264±0.001**	**0.170**	**0.264**	0.177	0.273	0.173	0.267	0.178	0.269	0.190	0.279	0.214	0.295	0.185	0.278	0.209	0.301
	720	**0.208±0.001**	**0.297±0.001**	0.212	0.299	0.213	0.304	0.210	0.309	0.225	0.317	0.230	0.313	0.255	0.327	0.220	0.315	0.245	0.333
	avg	**0.167±0.001**	0.260±0.001	0.168	**0.259**	0.175	0.269	0.173	0.268	0.178	0.270	0.189	0.276	0.217	0.295	0.189	0.278	0.212	0.300
Traffic	96	0.432±0.002	0.279±0.001	0.458	0.296	0.406	0.277	0.430	0.294	**0.395**	**0.268**	0.427	0.272	0.651	0.391	0.593	0.378	0.650	0.396
	192	0.442±0.001	0.289±0.001	0.457	0.294	0.426	0.290	0.452	0.307	**0.417**	**0.276**	0.454	0.289	0.602	0.363	0.595	0.377	0.598	0.370
	336	0.456±0.002	0.295±0.002	0.470	0.299	**0.432**	**0.281**	0.470	0.316	0.433	0.283	0.450	0.282	0.609	0.366	0.609	0.385	0.605	0.373
	720	0.487±0.003	0.311±0.001	0.502	0.314	**0.463**	**0.300**	0.498	0.323	0.467	0.302	0.484	0.301	0.647	0.385	0.673	0.418	0.645	0.394
	avg	0.454±0.002	0.293±0.001	0.472	0.301	0.431	0.287	0.463	0.310	**0.428**	**0.282**	0.454	0.286	0.627	0.376	0.618	0.390	0.625	0.383

Ablation study

FAIR Lab

KWANGWOON
UNIVERSITY

Table 11: Ablation study results of key modules. The best results are in **bold**.

Dataset		ETTh1		ETTm2		Weather		Electricity	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TimeEmb	96	**0.366**	**0.387**	**0.163**	**0.242**	0.150	**0.190**	**0.136**	**0.231**
	192	**0.417**	**0.416**	**0.226**	**0.285**	0.200	**0.238**	**0.153**	**0.246**
	336	0.457	**0.436**	**0.286**	**0.324**	0.259	**0.282**	**0.170**	**0.264**
	720	**0.459**	**0.460**	**0.383**	**0.381**	0.339	**0.336**	**0.208**	**0.297**
	avg	**0.425**	**0.425**	**0.265**	**0.308**	0.237	**0.262**	**0.167**	**0.260**
Random	96	0.407	0.415	0.240	0.318	0.197	0.237	0.242	0.344
	192	0.453	0.443	0.299	0.351	0.249	0.279	0.251	0.354
	336	0.499	0.471	0.361	0.386	0.314	0.321	0.291	0.393
	720	0.550	0.524	0.452	0.434	0.422	0.384	0.370	0.455
	avg	0.477	0.463	0.338	0.372	0.296	0.305	0.289	0.387
Zero	96	0.405	0.414	0.240	0.317	0.198	0.237	0.239	0.342
	192	0.452	0.442	0.299	0.351	0.248	0.279	0.248	0.352
	336	0.497	0.470	0.360	0.385	0.314	0.321	0.289	0.391
	720	0.547	0.523	0.450	0.433	0.421	0.383	0.367	0.453
	avg	0.475	0.462	0.337	0.372	0.295	0.305	0.286	0.385
Mean	96	0.397	0.410	0.204	0.289	0.175	0.221	0.241	0.339
	192	0.444	0.439	0.266	0.327	0.220	0.261	0.250	0.349
	336	0.488	0.464	0.328	0.364	0.267	0.296	0.291	0.387
	720	0.537	0.517	0.427	0.418	0.346	0.348	0.374	0.453
	avg	0.467	0.458	0.306	0.350	0.252	0.282	0.289	0.382
w/o. $\boldsymbol{X}_s$	96	0.376	0.391	0.173	0.252	0.182	0.221	0.178	0.259
	192	0.431	0.422	0.237	0.294	0.228	0.259	0.184	0.266
	336	0.470	0.441	0.295	0.332	0.282	0.299	0.200	0.282
	720	0.487	0.471	0.391	0.389	0.356	0.347	0.241	0.316
	avg	0.441	0.431	0.274	0.317	0.262	0.282	0.201	0.281
w/o. $\mathcal{H}_\omega$	96	**0.366**	0.391	0.164	0.244	0.151	0.192	0.137	0.233
	192	**0.417**	0.421	0.229	0.287	0.201	0.239	0.155	0.249
	336	**0.451**	0.441	**0.286**	**0.324**	0.259	0.283	0.172	0.269
	720	0.492	0.474	0.385	0.382	0.341	0.337	0.210	0.299
	avg	0.432	0.432	0.266	0.309	0.238	0.263	0.169	0.263
w/o. RevIN	96	0.383	0.403	0.193	0.286	**0.147**	0.191	0.137	0.235
	192	0.450	0.449	0.283	0.352	**0.194**	**0.238**	0.154	0.252
	336	0.485	0.463	0.427	0.449	**0.248**	0.290	0.171	0.271
	720	0.517	0.511	0.516	0.482	**0.326**	0.346	0.213	0.306
	avg	0.459	0.457	0.355	0.392	**0.229**	0.266	0.169	0.266

Framework Performance

Table 8: We equip Fredformer and CycleNet with TimeEmb on ETTh2. It brings stable performance(MSE) gains with trivial extra training costs.

Horizon	96	192	336	720
Fredformer	0.293	0.371	0.382	0.415
+TimeEmb	**0.289**	**0.358**	**0.360**	**0.387**
Impr.	1.4%	3.5%	5.8%	6.7%
former param	32820131	33465731	9894911	13656959
current param	32830860	33476460	9905640	13667688
extra param	**10729 (0.03%-0.11%)**			
CycleNet	0.285	0.373	0.421	0.453
+TimeEmb	**0.277**	**0.351**	**0.399**	**0.415**
Impr.	2.8%	5.9%	5.2%	8.4%
former param	99080	148328	222200	419192
current param	109809	159057	232929	429921
extra param	**10729 (2.56%-10.83%)**			

이후 계획

- Time-Frequency Pattern-Specific (TFPS)
  - 개선점 도출 및 추가 실험
- TimeEmb
  - 재현 실험 및 아이디어

부록